

$\mathbb{R}^{2k}$ is Theoretically Large Enough for Embedding-based Top- k Retrieval

Minimal Embeddable Dimension

Zihao Wang Hang Yin Lihui Liu Hanghang Tong Yangqiu Song Ginny Wong
Simon See

ICML 2026

Geometric approximability is not the obstruction.

The approximability question

Embedding retrieval stores objects as vectors, embeds a query, and retrieves by score comparison.

Minimal Embeddable Dimension

Given m objects, MED is the least d such that some vectors in $\mathbb{R}^d$ can exactly retrieve every answer set

$$S \subseteq X, \quad 1 \leq |S| \leq k$$

by score comparison and a threshold.

This isolates exact geometric capacity from positive-margin or learned embedding protocols.

Exact capacity is $\Theta(k)$, not a function of m

Scoring rule	Lower bound	Upper bound
Inner product	$k - 1$	$2k$
Euclidean distance	$k - 1$	$2k$
Cosine similarity	$k - 1$	$2k + 1$

- Lower bounds come from VC dimension.
- Upper bounds are explicit constructions.
- Exact geometric capacity does not grow with the universe size m .

The construction in one line

Place objects on the moment curve:

$$v_i = (t_i, t_i^2, \dots, t_i^{2k}).$$

For a target subset S , form

$$P_S(t) = \prod_{i \in S} (t - t_i).$$

Use the coefficients of $-P_S(t)^2$ as the query vector.

Closed-form witness

Selected objects score at the shared maximum; every other object scores lower.

Robust MED: normalized gaps change the regime

Exact MED is the no-margin problem.

RMED adds unit-norm vectors and a fixed selected-unselected score gap ϵ .

Positive gap makes m reappear:

$$\text{RMED}(m, k, \epsilon) \geq \frac{\log \binom{m}{k}}{\log(1 + 2/\epsilon)}.$$

Score gaps above

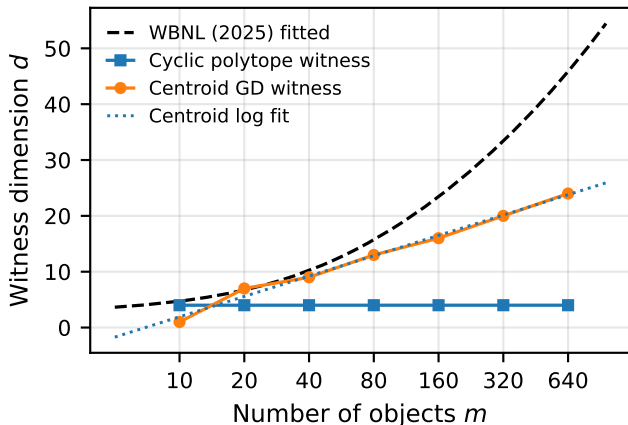
$$\epsilon_{\star}(m, k) = \frac{m}{\sqrt{k(m-1)(m-k)}} \sim \frac{1}{\sqrt{k}}$$

are infeasible when $m \gg k$.

At the feasible scale, a Gaussian centroid witness gives

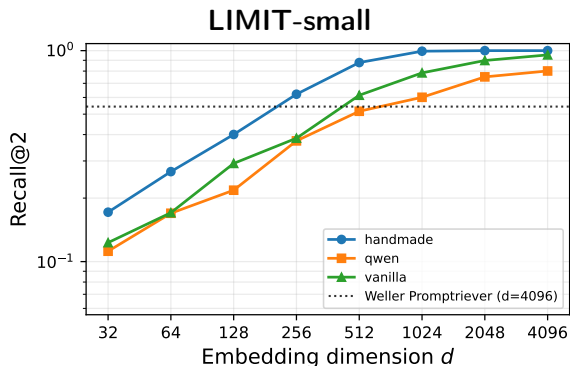
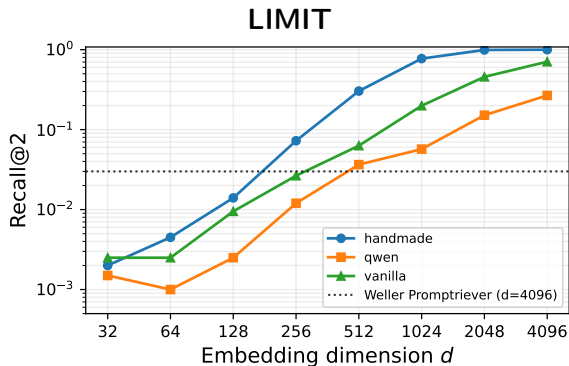
$$\text{RMED-C}(m, k, c/\sqrt{k}) \leq O(k^2 \log m).$$

Synthetic top-2: checked witnesses, not lower bounds



- Cyclic polytope gives an exact $d = 4$ witness for every tested m .
- Centroid GD and its log fit stay far below the WBNL fitted curve.
- A failed optimization run is not a proof of infeasibility.

LIMIT: simple single-vector baselines are strong



All three tokenizations cross the reported Promptriever line; vanilla is label-unaware and unsupervised. The packaged LIMIT top-2 tasks can also be exactly overfit in $\mathbb{R}^4$ by the cyclic-polytope construction.

What this means for retrieval

Exact MED

Cyclic-polytopal neighborliness gives $\Theta(k)$ dimensions for inner product, cosine, and Euclidean scoring, independent of m up to constants.

Robust MED

The normalized-gap problem is different: m reappears, and Gaussian centroid witnesses give logarithmic growth in m in the feasible regime.

Empirical interpretation

The synthetic and LIMIT failures are not failures of geometric approximability; they are entangled with optimization protocol, tokenization, or neural embedding models.